



Ramakrishna Mission Vivekananda Educational and Research Institute

PO Belur Math, Howrah, West Bengal 711 202

School of Mathematical Sciences

Department of Computer Science

MSc BDA/CS : Batch 2024-26, Semester I, Mid Exam

DA102: Basic Statistics

Dr. Sudipta Das

Student Name (in block letters):

Date: 14 September 2024

Student Roll No:

Max Marks: 60

Signature:

Time: 2hrs

Answers must be properly justified to deserve full credits.

1. (20 points)

The following data give the time in months from hire to promotion to manager for a random sample of 25 software engineers from all software engineers employed by a large telecommunications firm.

5	7	229	453	12	14	18	14	14
483	22	21	25	23	24	34	37	34
49	65	47	67	69	192	125		

(a) (10 points) Calculate the mean, median, mode, coefficient of variation and inter-quartile range for this data sample.

(b) (10 points) Draw a box plot and check for any outliers in this data sample.

2. (10 points)

From a sample of observations the arithmetic mean and variance are calculated. It is then found that one of the values, x_1 , is in error and should be replaced by x'_1 . Show that the adjustment to the variance to correct this error is

$$\frac{1}{n}(x'_1 - x_1) \left(x'_1 + x_1 - \frac{x'_1 - x_1 + 2T}{n} \right),$$

where $T = \sum_{i=1}^n x_i$

3. (15 points)

For a series data $\{x_1, \dots, x_n\}$, prove that

$$\frac{R^2}{2n} \leq \sigma^2 \leq \frac{R^2}{4},$$

where σ^2 and R are the variance and the range of the data, respectively.

4. (15 points)

The production of steel in seven different years are reported as follows

Years (X)	1951	1952	1953	1954	1955	1956	1957
Production in tons (Y)	201	263	314	395	427	504	612

From this record calculate the following

(a) (3 points) Draw a scatter plot.

(b) (12 points) Fit a best curve of the form

$$Y = ab^{(X-c)}.$$

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This exam has total 4 questions, for a total of 60 points and 0 bonus points.
